

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Faculty of Engineering and Architecture

2. Subject Name: Geotechnical Engineering
Credits: 4

L-T-P: 4-0-0

3. Pre-requisites: Solid Mechanics

4. Syllabus and reference books:

Syllabus:

Introduction: formation of soils, index properties and classification of soils, soil mineralogy.

Soil compaction, Standard Proctor and Modified Proctor compaction, field control of compaction: nuclear and electrical methods.

Darcy's law. Permeability of soils, methods of determining coefficient of permeability, Seepage analysis, flow nets, piping, quick sand condition.

Effective stress, soil-water-system, surface tension and capillarity.

Computing stresses within soil layers by the Boussinesq and Newmark approaches. Compressibility and consolidation of soils, normally and over consolidated soil, determination of over consolidation pressure, time rate of consolidation, determination of coefficient of consolidation, primary and secondary consolidation settlements.

Shear strength of soils: impacts of drained and undrained behavior, dilatancy, and partial saturation; direct shear and tri-axial tests, Mohr-Coulomb strength criterion, drained, consolidated undrained and drained test, pore pressures, Skempton's pore pressure coefficients, stress path.

Earth pressure theories, smooth and rough walls, inclined backfill and depth of tension crack.

Reference Books:

- 1) Smith I. Smith's Elements of Soil Mechanics, 9th Edition, Willey Blackwell
- 2) Das B. M. Principles of Geotechnical Engineering, 7th Edition, Cengage Learning
- 3) Das B.M. Advanced Soil Mechanics, 5th Edition, CRC Press, (ebook in Taylor and Francis)
- 4) Knappett J. and Craig RF. Craig's Soil Mechanics. 9th Edition, CRC Press (A Taylor & Francis ebook.)
- 5) Lambe T. W. and Whitman R. V. Soil Mechanics, Willey

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Introduction: formation of soils, index properties and classification of soils, soil mineralogy.	6
2.	Soil compaction, Standard Proctor and Modified Proctor compaction, field control of compaction: nuclear and electrical methods.	4
3.	Darcy's law. Permeability of soils, methods of determining coefficient of permeability, Seepage analysis, flow nets, piping, quick sand condition.	6
4.	Effective stress, soil-water-system, surface tension and capillarity.	4
5.	Computing stresses within soil layers by the Boussinesq and Newmark approaches.	4
6.	Compressibility and consolidation of soils, normally and over consolidated soil, determination of over consolidation pressure, time rate of consolidation, determination of coefficient of consolidation, primary and secondary consolidation settlements.	8
7.	Shear strength of soils: impacts of drained and undrained behavior, dilatancy, and partial saturation; direct shear and tri-axial tests, Mohr-Coulomb strength criterion, drained, consolidated undrained and drained test, pore pressures, Skempton's pore pressure coefficients, stress path.	14
8.	Earth pressure theories, smooth and rough walls, inclined backfill and depth of tension crack.	6
Total number of hours		52